

NAME:

MATH 316 – HW 8

Due: Tuesday Oct 20, 2026

HOMEWORK DAY 24 – *Step functions, functions defined piecewise. Heaviside function. §3.1*

1. (p157:6e) Solve the following IVP with the Laplace transform method

$$x'' - 2x' + 2x = e^{-t}, \quad x(0) = 0, \quad x'(0) = 1$$

2. (a) State the definition of $H(t)$ and $H(t - a)$, $a > 0$. Graph each function.

(b) Use the definition of the Laplace transform to find $\mathcal{L}[H(t - a)]$.

3. Consider the function $f(t) = \begin{cases} 0, & 0 \leq t < 3 \\ -2, & 3 \leq t < 5 \\ 2, & 5 \leq t < 7 \\ 1, & 7 \leq t \end{cases}$

(a) Sketch the graph of f

(b) Write f in terms of the Heaviside function

(c) Find $L[f(t)]$

4. Let $f(t) = t^2$.

(a) Plot $f(t)$

(b) Plot $f(t)H(t-3)$

(c) Plot $f(t-3)$

(d) Plot $f(t-3)H(t-3)$

5. Use the definition of the Laplace transform to find $\mathcal{L}[f(t-a)H(t-a)]$.
Write the answer in terms of $L[f(t)] = F(s)$.

6. Consider the function

$$f(t) = \begin{cases} 4 & , \quad 0 \leq t < 2 \\ t^2 & , \quad 2 \leq t \end{cases}$$

- (a) Sketch the graph of f

- (b) Write f in terms of the Heaviside function

- (c) Find $L[f(t)]$

7. Find the following Laplace Transforms.

(a) $\mathcal{L}[e^{-2t}H(t-3)]$

(b) $\mathcal{L}[tH(t-4)]$

(c) $\mathcal{L}[\cos(t-\pi)H(t-\pi)]$ (p145: 8f)

8. Find the following inverse Laplace Transforms.

(a) $\mathcal{L}^{-1}\left[e^{-4s}\frac{7}{s}\right]$ (p145: 9d)

$$(b) \quad \mathcal{L}^{-1} \left[e^{-s} \frac{1}{s(s-2)} \right] \quad (\text{p145: 9e})$$

$$(c) \quad \mathcal{L}^{-1} \left[e^{-\pi s} \frac{3s}{s^2+9} \right] \quad (\text{p145: 9h})$$

$$(d) \quad \mathcal{L}^{-1} \left[\frac{1}{s^2} - e^{-2s} \frac{1}{s^2} \right] \quad (\text{p156: 3k})$$

9. Consider the following IVP

$$x' + 5x = H(t - 2) , \quad x(0) = 1$$

(a) Solve the IVP in two pieces (first on $t \in [0, 2]$, then on $t \in [2, \infty)$) using the method of undetermined coefficients.

(b) Solve the IVP using the Laplace Transform.
Write your answer as a function defined piecewise.

10. Solve the following IVP using the Laplace Transform.
Write your answer as a function defined piecewise.

$$x'' - x = \begin{cases} t & , \quad t < 1 \\ 0 & , \quad t \geq 1 \end{cases} , \quad x(0) = 0, \quad x'(0) = 0.$$

11. Solve the following IVP using the Laplace Transform. Write your answer as a function defined piecewise.

$$x'' + 4x = f(t) , \quad x(0) = 0 , \quad x'(0) = 1$$

$$\text{where } f(t) = \begin{cases} 3 & , \quad t < 1 \\ 5 & , \quad 1 \leq t < 5 \\ -2 & , \quad 5 \leq t \end{cases}$$

(Ans: $x(t) = \frac{1}{2} \sin(2t) + 3g(t) + 2H(t-1)g(t-1) - 7H(t-5)g(t-5)$, where $g(t) = \frac{1}{4}(1 - \cos(2t))$.
Write it piecewise.)

12. Find the inverse Laplace transform of

$$F(s) = \frac{a}{s^2(s^2 + a^2)}.$$

Your answer may not be in terms of a convolution integral.

13. $x'' + 4x = g(t)$, $x(0) = 3, x'(0) = 0$. You may leave your solution in terms of a convolution integral.

14. One can show with convolution that

$$\mathcal{L}^{-1}\left[\frac{1}{(s^2 + k^2)^2}\right] = \frac{1}{2k^3}[\sin(kt) - kt \cos(kt)] .$$

Use this result to solve

$$x'' + 9x = \sin(3t), \quad x(0) = x'(0) = 0.$$

Is there a relation between your solution and the phenomenon of resonance discussed earlier in this class?

15. Compute $\int_0^\infty e^{-2(t-3)^2} \delta(t-4) dt$.

Solve the following initial value problems. In all cases, write your answer as a function defined piecewise.

16. $x' + 3x = \delta(t-1) + H(t-4)$, $x(0) = 1$

17. $x'' - x = \delta(t - 5)$, $x(0) = x'(0) = 0$. Also sketch the solution.

18. $x'' + 4x = \delta(t - 2) - \delta(t - 5)$, $x(0) = 0$, $x'(0) = 1$. Also sketch the solution.